

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Experiment No.: PBL Q.4 (CO-1)

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1. TITLE

Document Object Model (DOM) and Dynamic Modification of Content, Style and Attributes

2. AIM

To explain the Document Object Model (DOM) and to demonstrate how JavaScript can dynamically change the content, style and attributes of an HTML element.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the structure of the DOM.
- To access HTML elements using JavaScript.
- To change the content of an element dynamically.
- To change the style of an element dynamically.
- To change the attributes of an element dynamically.
- To handle user actions using event listeners.

4. THEORY

What is the DOM? The Document Object Model (DOM) is a programming interface for HTML documents. When a browser loads a web page, it converts the HTML into a tree of objects called nodes. JavaScript can use this tree to read, change, add or remove any part of the page while it is displayed, without reloading it. This is what makes web pages dynamic.

DOM Tree Structure. The document is the root of the tree. Every HTML tag becomes an element node, the text inside a tag becomes a text node, and each attribute becomes an attribute node. Nodes are related as parent, child and sibling. The DOM tree of the demonstration page is:

```
document
  |-- html
    |-- head
      |   |-- meta
      |   |-- title --> "DOM Demonstration"
    |-- body
      |-- div#box
        |   |-- h1#heading --> "Original Heading"
        |   |-- p#para --> "This is the original paragraph text."
        |   |-- a#link [href = "https://www.example.com"]
        |   |-- input#nameField [placeholder = "Original placeholder"]
      |-- button (x4)
      |-- div#info
```

Types of nodes. Document node (the whole page), Element nodes (e.g. <h1>, <p>), Attribute nodes (e.g. href, placeholder) and Text nodes (the text inside elements).

Accessing elements. document.getElementById("id"), document.querySelector("selector"), document.getElementsByClassName() and document.getElementsByTagName() are used to select elements.

Changing content, style and attributes:

What to change	DOM property / method
Content	element.textContent (plain text), element.innerHTML (text with HTML tags)
Style	element.style.property (e.g. style.color, style.backgroundColor, style.fontSize), element.classList
Attributes	element.setAttribute(name, value), element.getAttribute(name), element.removeAttribute(name)
Events	element.addEventListener("click", function)

The basic flow of the demonstration is:

User Click -> Event Listener -> JavaScript Function -> DOM Update -> Browser Re-renders the Page

5. ALGORITHM / PROCEDURE

1. Create an HTML page with a heading, paragraph, link and input field inside a container div.
2. Add four buttons: Change Content, Change Style, Change Attributes and Reset, and an info area.
3. Access all the elements in JavaScript using document.getElementById().
4. Write changeContent() to modify heading.textContent and para.innerHTML.
5. Write changeStyle() to modify box.style and heading.style properties.
6. Write changeAttributes() to modify the link and input using setAttribute() and display the values using getAttribute().
7. Write resetAll() to restore the original content, style and attributes.
8. Attach a click event listener to each button using addEventListener().
9. Open the page in the browser and click each button.
10. Observe the changes and verify the actual output with the expected output.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<!-- Program: DOM Demonstration
      Purpose: Show how JavaScript changes the content, style and attributes
              of HTML elements dynamically using the DOM. -->
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM Demonstration</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 30px; background: #f7f7f7; }
    /* Container box whose style will be changed by JavaScript */
    #box {
      background: #ffffff;
```

```

        border: 2px solid #999;
        padding: 20px;
        width: 520px;
        border-radius: 6px;
    }
    button { padding: 8px 12px; margin: 10px 6px 0 0; cursor: pointer; }
    /* Area that shows a message after every action */
    #info {
        margin-top: 16px;
        padding: 10px;
        width: 520px;
        background: #eef;
        font-family: monospace;
        font-size: 13px;
        box-sizing: border-box;
    }
</style>
</head>
<body>
    <!-- Elements that JavaScript will modify (each has an id) -->
    <div id="box">
        <h1 id="heading">Original Heading</h1>
        <p id="para">This is the original paragraph text.</p>
        <a id="link" href="https://www.example.com">Original link</a>
        <br><br>
        <input type="text" id="nameField" placeholder="Original placeholder">
    </div>

    <!-- Buttons that trigger the DOM changes -->
    <button id="btnContent">Change Content</button>
    <button id="btnStyle">Change Style</button>
    <button id="btnAttr">Change Attributes</button>
    <button id="btnReset">Reset</button>

    <div id="info">Click a button to modify the DOM.</div>

    <script>
        // Access the elements using the DOM (getElementById returns the element)
        const box      = document.getElementById("box");
        const heading   = document.getElementById("heading");
        const para      = document.getElementById("para");
        const link      = document.getElementById("link");
        const field     = document.getElementById("nameField");
        const info      = document.getElementById("info");

        // 1. Change CONTENT
        function changeContent() {
            // textContent: sets plain text only
            heading.textContent = "Heading Changed by JavaScript";
            // innerHTML: sets text that can contain HTML tags (<b>, <i>)
            para.innerHTML = "The paragraph now uses <b>innerHTML</b> " +
                "to add <i>formatted</i> text.";
            link.textContent = "Visit MDN Web Docs";
            info.textContent = "Content changed using textContent and innerHTML.";
        }

        // 2. Change STYLE
        function changeStyle() {
            // element.style.property changes the inline CSS of the element

```

```

        box.style.backgroundColor = "#fff3cd";
        box.style.border = "3px dashed #d35400";
        heading.style.color = "#c0392b";
        heading.style.fontSize = "36px";
        para.style.fontStyle = "italic";
        info.textContent = "Style changed using the element.style property.";
    }

    // 3. Change ATTRIBUTES
    function changeAttributes() {
        // setAttribute(name, value) adds or changes an attribute
        link.setAttribute("href", "https://developer.mozilla.org");
        link.setAttribute("title", "Mozilla Developer Network");
        link.setAttribute("target", "_blank");
        field.setAttribute("placeholder",
            "Placeholder changed by setAttribute()");
        field.setAttribute("maxlength", "20");
        // getAttribute(name) reads the new values, shown in the info box
        info.textContent = "href = " + link.getAttribute("href") +
            " | title = " + link.getAttribute("title") +
            " | maxlength = " + field.getAttribute("maxlength");
    }

    // 4. Reset everything to the original state
    function resetAll() {
        heading.textContent = "Original Heading";
        para.textContent = "This is the original paragraph text.";
        link.textContent = "Original link";
        // removeAttribute("style") removes all inline styles we added
        box.removeAttribute("style");
        heading.removeAttribute("style");
        para.removeAttribute("style");
        link.setAttribute("href", "https://www.example.com");
        link.removeAttribute("title");
        link.removeAttribute("target");
        field.setAttribute("placeholder", "Original placeholder");
        field.removeAttribute("maxlength");
        info.textContent = "DOM reset to original state.";
    }

    // Event listeners: run a function when its button is clicked
    const byId = (id) => document.getElementById(id);
    byId("btnContent").addEventListener("click", changeContent);
    byId("btnStyle").addEventListener("click", changeStyle);
    byId("btnAttr").addEventListener("click", changeAttributes);
    byId("btnReset").addEventListener("click", resetAll);
</script>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<code>document.getElementById()</code>	Selects an element from the DOM using its id.
<code>textContent</code>	Changes the plain text of an element.
<code>innerHTML</code>	Changes the content of an element, including HTML tags.
<code>element.style.color</code>	Changes an inline CSS property of the element.

Code	Explanation
setAttribute(name, value)	Adds or changes an attribute (href, title, placeholder, ...).
getAttribute(name)	Reads the current value of an attribute.
removeAttribute(name)	Removes an attribute (used to reset the page).
addEventListener("click", fn)	Runs the function when the button is clicked.

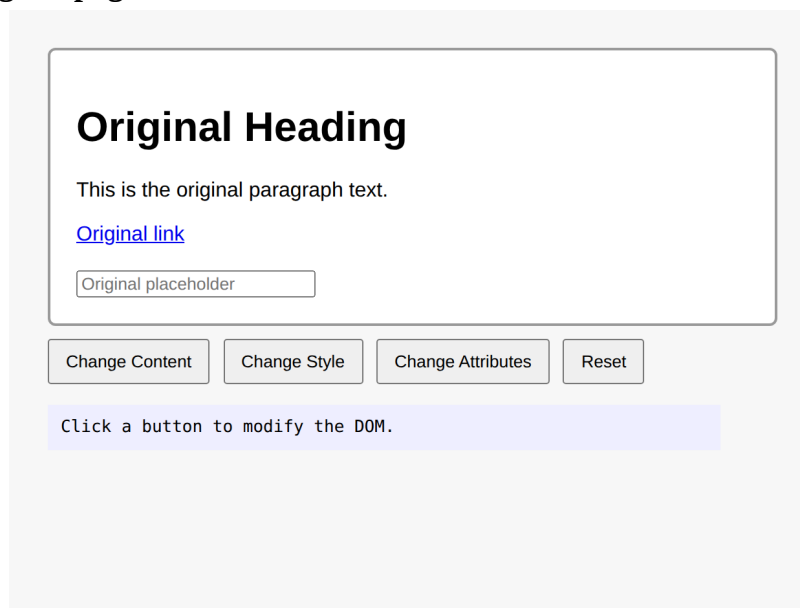
8. TEST CASES AND EXECUTION DATA

Each button was clicked in the browser and the resulting DOM values were recorded.

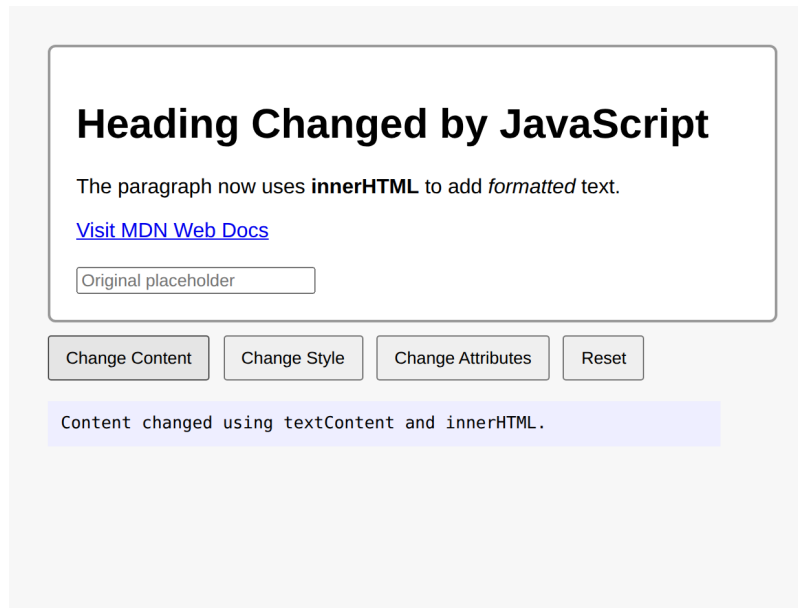
Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open the page	Original heading, paragraph, link and placeholder are shown	Heading: "Original Heading"; link: "Original link"	Pass
TC-02	Click "Change Content"	Heading text changes; paragraph shows bold and italic text (innerHTML); link text changes	Heading: "Heading Changed by JavaScript"; paragraph HTML contains and <i>; link: "Visit MDN Web Docs"	Pass
TC-03	Click "Change Style"	Background #fff3cd, dashed border, heading red (#c0392b) and 36px, paragraph italic	Background rgb(255, 243, 205); border dashed; heading rgb(192, 57, 43), 36px; paragraph italic	Pass
TC-04	Click "Change Attributes"	href, title, target of the link and placeholder, maxlength of the input are changed	href=https://developer.mozilla.org; title="Mozilla Developer Network"; target=_blank; maxlength=20; placeholder changed	Pass
TC-05	Click "Reset"	All content, styles and attributes return to the original state	Heading: "Original Heading"; background rgb(255, 255, 255); href=https://www.example.com; title=null	Pass

9. OUTPUT

Output 1: Original page



Output 2: After clicking "Change Content"



Output 3: After clicking "Change Style"



Output 4: After clicking "Change Attributes"



Output 5: After clicking "Reset"



10. RESULT

Thus, the Document Object Model was explained, and it was demonstrated that JavaScript can dynamically change the content, style and attributes of HTML elements. The behaviour was verified using multiple test cases.